



Cartography II: Collective cartographies in the social media era

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Abstract

The goal of this second report is to review how social media are changing the way we collectively map the world. To reach this goal I review different collective mapping practices that characterize the social media era. First I examine the situation of community mapping in the context of new cartographic processes and technologies, with a focus on indigenous cartographies. I then review the use of volunteers in the production and representation of geospatial knowledge, with an emphasis on crisis mapping. Finally, I discuss how map-making in the social media era reflects major trends in terms of power relationships that occur between the state, its citizens and the private sector. These trends reveal the replacement of the state as the main reference for the collection and dissemination of cartographic data, by a combination of private interest and individually volunteered contributions. Just as the specific interests of the nation state have largely helped to shape the reality produced by paper maps throughout the centuries, this new convergence of interests is now helping to shape the reality produced by digital maps through geosocial media.

Keywords

community cartography, crisis mapping, Google map, OpenStreetMap, social media

1 Introduction

On 13 December 2010, Paul Butler, an intern at Facebook, released a world map of Facebook friendships that received much attention across social media. This map was elected the best map of 2011 by Spatial Analysis (2011) and inspired the development of other similar visualizations such as the ‘Map of scientific collaborations from 2005 to 2009’ (Beauchesne, 2011). Butler’s map represented ‘a sample of about ten million pairs of friends’ connected with bright lines on a black background map of the world (Butler, 2010). While Butler was struck by the fact that the results represented ‘real human relationships’, what seems even more striking is the overall message the map conveys. Not

only does it materialize the global empire that Facebook has been building over the past few years (Joliveau, 2011), which follows the geography of internet users mapped more than a decade ago (Dodge and Kitchin, 2001), but it also associates this empire with enlightenment. The parts of the globe with many Facebook members (mainly in the western world) appear in bright white, while the rest of the world (including Russia, China, the Middle East and most of Africa) remains dark. Although

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this map could be considered as simply one more cartographic avatar contributing to the reinforcement of western hegemony through access to technology, it also illustrates some of the major trends that cartography is experiencing within the context of social media. I propose to review these trends in this second report, focusing on collective cartographic practices.

Enabled by Web 2.0, personal contributions to collective knowledge through social media applications such as Facebook, Twitter, Wikipedia and Foursquare are now systematically labelled with geographic coordinates. As a result, this geotagged data can be easily retrieved and mapped using online mapping platforms such as Google Maps and OpenStreetMap, as well as a growing number of ad hoc cartographic applications (Beauchesne, 2011; Butler, 2010; Elwood and Leszczynski, 2011; Graham and Zook, 2011; Kelley, 2013). The cartography of this collective knowledge, made of geographic layers, personal information and collective stories, contributes to the reshaping of places and communities. In light of this context, this report explores the new collective cartographic practices enabled by social media. It focuses first on community mapping, specifically indigenous cartographies, then on collaborative mapping and crowdsourcing with an emphasis on crisis mapping. Finally, the changing relationships between the state, its citizens and the private sector in this context are reviewed and discussed.

II Community mapping

Historically, one of the main forms of community mapping has been 'indigenous cartography' (Parker, 2006). Since the 1960s, indigenous groups have produced a range of spatial expressions through dance, song, painting, dreaming and mapping, to reclaim indigenous sovereignty over lands, to negotiate aboriginal rights, and to regain dignity during conflicts with governments and institutions (Rundstrom, 1991, 1995; Sparke, 1998; Woodward and Lewis, 1998). The political

function of these indigenous cartographies remains at the centre of recent research on community mapping (Beyersdorff, 2007; Chapin et al., 2005; Hirt, 2012; Louis et al., 2012; Pearce and Louis, 2008; Roberts, 2012; Vermeyley et al., 2012). Although most of this research focuses on deconstructing the map and mapping processes as evidence of historical and cultural rights against governmental and institutional infringements, other types of emotional and internal power dynamics can be revealed (Kitchin et al., 2013; Young and Gilmore, 2013). For instance, through an in-depth study of participatory mapping processes in Trinidad and in Venezuela, Sletto (2009) unveils the power structures that exist within some indigenous communities, emphasizing the performative dimension of community mapping as well as the consequences that such processes can have on the overall structure of a community.

Since the 1990s, indigenous cartographic processes and practices have been affected by digital media, as illustrated by the extensive use of participatory GIS (PGIS) in North American indigenous communities (Chapin et al., 2005; Dunn, 2007). While these practices are often strongly criticized because of the subordination of indigenous spatial world-views to western technologies and perspectives (Louis et al., 2012; Turnbull, 2008; Vermeyley et al., 2012; Wainwright and Bryan, 2009; Wood, 2010c), some recent examples illustrate how certain indigenous groups are taking advantage of geospatial technologies to push forward their political agendas. In Canada, for instance, certain indigenous communities are working with research institutions, developing collaborative online cartographic applications to take ownership of their projects (Brauen et al., 2011; Caquard et al., 2009; Eisner et al., 2012; Laidler et al., 2011; Pulsifer et al., 2011; Pyne and Taylor, 2012). In what might be viewed as progress over previous PGIS practices, many of these projects explore the possibilities of combining indigenous and scientific spatial knowledge to develop

hybridized forms of spatial representation that recognize and respect the uniqueness and importance of indigenous spatial expressions (Brigg and Maddison, 2012; Laituri, 2011; Palmer, 2009, 2012; Pearce and Hermann, 2010; Pearce and Louis, 2008; Pyne and Taylor, 2012). These hybrid cartographic forms of expressions do not reverse colonial social relations, but rather rework them (Wainwright and Bryan, 2009), helping to develop a new space of mutual understanding, provided that the balance between western science and indigenous knowledge is respected (Turnbull, 2009).

Community mapping has also emerged in other settings. For instance, parish maps have been collectively designed in the UK since 1985 to support local distinctiveness and local empowerment (Crouch and Matless, 1996; Burini, 2012; Perkins, 2007; Wood, 2010c). Contemporary artists have also developed different kinds of collaborative cartographic projects and mapping performances. Most of these works have been designed to enable local communities and marginalized groups to express their own perspectives on their territories through alternative cartographic processes. While these artists do not necessarily rely on Web 2.0 to achieve their goals (Cassidy, 2012; Kanarinka, 2011; Kanouse, 2011; Nold, 2009; Wood, 2010a), many are exploring its participatory potential. This is, for instance, the case with Jake Barton, who collects and maps personal memories of New York City (Krygier, 2006; Wood, 2010b, 2010c). Barton's *City of Memory* project has been defined as a 'Web-based model for Participatory GIS' (PGIS) since it is truly participatory, public and geographic, and generates all kinds of information from citizens, which is not often the case with more conventional PGIS projects (Wood, 2010c).

III Collaborative mapping

With geosocial media, more and more citizens are contributing widely to the collective

production of spatial knowledge. This 'geocrowdsourcing' (Dodge and Kitchin, 2013) can be done voluntarily, as captured by the now well-used acronym VGI (volunteered geographic information) coined by Goodchild (2007), or involuntarily, as captured by the acronym iVGI (involunteered geographic information) (Fischer, 2012), through the recording of locations and activities by personal mobile devices (e.g. cell phones) and external devices (e.g. satellites) (Dodge and Perkins, 2009; Elwood and Leszczynski, 2011). This distinction is synthesized by Harvey (2013) with the terms 'opt-in' (volunteered) and 'opt-out' (contributed).

The volunteered participation is made possible by Web 2.0 mapping technologies and applications that support the simultaneous editing of content by multiple users (Dodge and Kitchin, 2013; Haklay et al., 2008; Kitchin and Dodge, 2011; Miller, 2006; Sui, 2008). These capabilities, combined with decreased costs in data storage (Sui et al., 2013) and with the development of infrastructures such as communication networks (O'Brien and Field, 2012) and data centres (Peterson, 2012), allow individuals to easily produce and access information from the ground with their own mobile devices (Wilson, 2011a). Although little is certain about why citizens voluntarily contribute geographical data (Craglia et al., 2008; Elwood et al., 2012), some authors explain this increasing involvement by referring to different personal motives such as idealism or local needs (Grira et al., 2009; Wilson, 2011b). Companies such as Google have played on this collective commitment to stimulate the participation of volunteers in their corporate project (Boulton, 2010; Dodge and Kitchin, 2013; Gerlach, 2010). For instance, with the Google Map Maker application, any registered individual can contribute voluntarily to the improvement and updating of Google Maps.

Although these new cartographic capabilities are driven by consumerist oriented applications

– as illustrated by the very first map mash-up that was designed for house hunting (Crampton, 2010) – they are also being used to develop new participatory mapping tools applied to different domains such as assisting park management (Elsley and Cartwright, 2011) and supporting decision-making processes (Cai and Yu, 2009; Rinner et al., 2008). According to Miller (2006), the very first real Participatory GIS in the Web 2.0 era might have been the first collective Google map mash-up created by volunteers in August 2005 to allow citizens in New Orleans (USA) to share information during the post-Hurricane Katrina crisis. This example not only emphasizes the capacity of geosocial media to support collective endeavour in a crisis situation, but also demonstrates its incapacity to seriously reduce social and historical divides. Indeed, an analysis of the geospatial content uploaded during the post-Hurricane Katrina crisis shows that ‘neighborhoods with high percentages of African Americans were significantly less likely to have informational comments about them posted’ on the map mash-up (Crutcher and Zook, 2009: 532).

This first collective map mashup was followed by many others in the emerging cross-disciplinary field of ‘crisis mapping’ (Liu and Palen, 2010; Meier, 2009a), as illustrated by the extensive number of Google MyMaps generated by volunteers during the Jesusita Fire in California in May 2009 (Goodchild and Glennon, 2010), as well as by the key role played by the Ushahidi collaborative mapping platform in Port-au-Prince after the Haiti earthquake in January 2010 (Heinzelman and Waters, 2010). As pointed out by Meier (2009b), Ushahidi is not only about collecting geospatial data from the crowd (i.e. ‘crowdsourcing’), but also about returning this information to the crowd (i.e. ‘crowdfearing’). This is where Thatcher (2013) makes the distinction between VGI and ‘Volunteer Information Services’ that correspond to the

use of VGI to serve the population by improving the coordination of actions in the field.

Crisis mapping raises different cartographic challenges such as how to provide a clear and unambiguous cartographic representation of a quickly changing situation on small-screen devices (Razikin et al., 2010), and how to help users to differentiate the origins and the quality of the data (e.g. authoritative or not, reliable or not) (Goodchild and Glennon, 2010; Roche et al., 2013). Multiple attempts have been made to measure and improve VGI quality for crisis mapping, including developing a workflow to improve the reliability of the data (De Longueville et al., 2010), formalizing a set of rules and procedures to ensure data consistency (Goodchild and Glennon, 2010: 240), and rating the quality of the contributors based on reputation systems such as eBay’s rating system (Maué, 2007; Seeger, 2008).

Data validation is key for the success of any collaborative mapping project, as illustrated by the amount of research done on this topic for the OpenStreetMap (OSM) project (Dodge and Kitchin, 2013; Girres and Touya, 2010; Haklay, 2010; Haklay et al., 2010; Lin, 2011; Mooney and Corcoran, 2012). The OSM project is often considered the most successful collective map ever produced (Gerlach, 2010; Johnson and Sieber, 2012; Perkins, 2007), relying on dedicated contributors to compile a publicly accessible road map for the entire world at a very fine scale. It has been described as a way to develop new forms of communities around the world, based on shared values and practices (Lin, 2011). The success of this citizen-driven endeavour has also attracted private interests (Caquard, 2013) and transformed its original amateur approach into professional and commercialized practices reworking its collaborative and egalitarian aspects (Budhathoki et al, 2010; Perkins, 2013; Haklay, 2013).

IV Maps, state, citizens and corporations in the geosocial media era

The professionalization and commercialization of the OSM project illustrates the new institutional regime that governs cartography within the Web 2.0 era. Although state agencies such as the Ordnance Survey in the UK, IGN in France and USGS in the USA remain the references for cartographic data, the increasing participation of citizens and the private sector signals the replacement of the state as the main authoritative reference for the collection and diffusion of geographic information (Leszczynski, 2012). In their inventory of 99 VGI initiatives, Elwood et al. (2012) found that almost two-thirds of these initiatives had been sponsored by for-profit institutions, with only 7% by governments. The state is now largely limited to being a regulatory body as well as an intermediary between its citizens and the private sector (Elwood and Leszczynski, 2011; Leszczynski, 2012). Although some authors (e.g. Dodge and Kitchin, 2013) argue that even crowdsourcing projects such as OSM reproduce the national cartographic and ontological standards determined by the state – and thus continue to ‘perform the state’ (Wood, 2012) – in some countries, such as China, VGI practices are seen as subtle ways of contesting state discourses (Lin, 2013).

From a governmental perspective, this connection between citizens and private corporations has several advantages. It can serve to promote active citizen participation in decision-making processes (Johnson and Sieber, 2012; Nuojuua, 2009; Rinner et al., 2008; Seeger, 2008) as in the case of political redistricting (Crampton, 2013), as well as improve the government’s efficiency (Craglia et al., 2008; Goodchild and Glennon, 2010) and its image of good governance (Johnson and Sieber, 2012). But it can also serve as a governmental strategy to distract citizens from other forms of democratic engagement (Johnson and

Sieber, 2013). It can also be seen as the characterization of the inherent incapacity of the state to fulfil some of its missions such as providing relevant geographic data to its citizens. This incapacity can be explained by the state’s historical reluctance to make all of its data publicly available (Johnson and Sieber, 2012), as well as by the neoliberal trend towards the reduction of governmental resources and responsibilities (Dodge and Kitchin, 2013; Wilson, 2011b).

Private-sector companies and citizens are now assuming some of these responsibilities. For instance, local residents in some communities in the USA are turning into ‘mapping bodies’, using handheld devices such as GPS and cell phones to collect and map data from their neighbourhoods, and to report issues and crime to the local authorities (Wilson, 2011b). As Wilson (2011b: 363) argues, this increasing responsibility of citizens is ‘in step with the devolution of state services’. In this emerging model, the citizen is increasingly addressed as a ‘prosumer’ (Ritzer and Jurgenson, 2010), both a consumer of spatial products at no cost and an unpaid producer of spatial data over the web (Dodge and Kitchin, 2013; Goodchild, 2007; Grira et al., 2009; Leszczynski, 2012). This represents a new corporate model of capital accumulation widely used in the context of the Web 2.0, which relies on unwaged labour and information resources (Ritzer and Jurgenson, 2010), as well as on the continuous development of new mapping applications.

Social media applications such as Foursquare and Gowalla, designed to provide insight about places based on comments and contributions from individuals (e.g. restaurants, stores, parks, schools), are now systematically linked to online mapping services such as 4sqmap, 4mapper or Gowalla Map, as well as to ‘meta-geosocial aggregators’ such as Fourwhere that give easy cartographic access to this collective knowledge (Kelley, 2013: 185). The local information collected through these applications can

be mapped to reveal spatial structures, patterns and social hotspots (Stefanidis et al., 2013). For instance, the application 'livehoods' (<http://www.livehoods.org/research>) was designed to redefine neighbourhood boundaries based on different comments associated with places via social media. Graham and Zook (2011) have also developed an application to aggregate and map user-generated Google placemarks. They argue that this geolocated, user-generated content is increasingly affecting our perceptions about places, since it produces a hierarchy of locations and sites based on the type and volume of associated user-generated content (e.g. pictures). Kelley (2013: 182) goes further by arguing that the data collected through geosocial media such as Foursquare 'can be assembled to speak to the imaginaries of sub-city scale communities' and to 'inform the ambient collective intelligence that structures how we come to know, experience and behave in particular places'.

Although geosocial media might be changing the way we view and interact with the world (Graham and Zook, 2011; Kelley, 2013), these changes do not equally concern all citizens. Haklay (2012) emphasizes with Budhathoki et al. (2010) that there is an over-representation of 'wealthy, powerful, educated and mostly male elite' in the digital representation of places, which leads him to seriously question 'the acceptance of the disproportional amount of information that these outliers produce as reality'. Just as the specific interests of the nation state have largely shaped the reality produced by paper maps throughout the centuries, the recent convergence of interests between high-tech private companies and a small group of technologically savvy individuals is now shaping the reality produced through geosocial media. In this sense, many collective mapping projects in the social media era can be seen as contributing to the reworking of existing power structures, rather than truly resisting them.

V Conclusion

Notions of community and citizenship are far from clearly defined, as illustrated by the diversity of perspectives compiled in a recent special issue of *Environment and Planning D* entitled 'Citizenship without community?' (Closs Stephens and Squire, 2012). The dualistic approach that has prevailed in community cartography (Perkins, 2007), envisioning communities as based either on 'territorial boundedness' or on 'sociocultural homogeneity' (Baker and Bartelson, 2009: 2), is no longer valid. Individuals are collaborating within as well as outside conventional communities to produce, map and retrieve collective knowledge. This collective mapping endeavour can have multiple positive implications as discussed throughout this report. It can serve as a base to develop new spaces of mutual understanding between communities, as illustrated in the context of indigenous cartographies. It can also be used to improve response time and efficiency during post-crisis situations, as illustrated in Haiti during the 2010 post-earthquake crisis. Overall, the most important promise of Web 2.0 mapping might lie 'in what it can tell about local activities in various geographic locations that go unnoticed by the world's media, and about life at a local level' (Goodchild, 2007: 220).

The reality of collective cartographies in the social media era has not yet fulfilled this promise. When there are obvious inequalities between different groups of people in terms of geospatial data contribution (as in the case of the African American community within post-Katrina New Orleans), then 'the resulting cyberscapes hide as much as they reveal' (Crutcher and Zook, 2009: 533). The cartographic content collectively produced via social media remains largely the expression of the values of a relatively small number of contributors with technological ability. In this sense, these collective cartographies are not that

different from early modern European maps which were described by Harley (1988: 70) as contributing to ‘the reproduction, the reinforcement, and the legitimation of cultural and political values’. The main difference is that for centuries these values had been defined by the state, while in the social media era they reflect the interests of an heterogeneous coalition of technologically savvy individuals and private-sector companies. Although these changes can have positive practical applications for a broad range of citizens, they may also indicate the emergence of a new form of corporate technocracy.

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